

**Project Refinement and Ethical Integration**  
**Inventory Organizer with AI-Assisted Administration**

Josue Neiculeo

Ensign College

IT 497: IT Capstone

## **1. The Original Project**

Earlier this semester I built an inventory management system for freelance clients, later adding an AI-assisted feature that flagged low-stock items and suggested reorder quantities. The original version worked, and clients found it useful, but it was built quickly and prioritized functionality over safeguards. It had no login system, no record of who changed what, and the AI's reorder suggestions were applied directly to the displayed recommendations without any structured review step. It also exported all inventory data, including client-identifying references, the same way every time, regardless of who was asking or why.

## **2. Complex Challenges Identified**

### ***Security***

The original version had no authentication at all — anyone with the link could view and, in some versions, edit inventory data. There was also no consistent use of parameterized queries in early drafts of the reorder logic, which is a known SQL-injection risk in any data-driven web application.

### ***Accountability and traceability***

Nothing in the original system recorded who made a change or when. If an item's stock count looked wrong, there was no way to trace what happened without asking the client directly and hoping they remembered.

### ***Scalability***

The dashboard loaded the entire inventory table at once. For a small client this was fine, but it would not hold up for a client with a large or growing catalog.

### ***User experience and accessibility***

The original interface used plain HTML with no labelled form fields, no status announcements, and no clear indication of low-stock items beyond a manually reviewed list. A non-technical user could miss something important.

### **3. Ethical Implications Identified**

#### ***Over-automation and job displacement***

This is the concern I have returned to throughout this course. An AI feature that quietly changes inventory counts on its own reduces the role of the person who used to do that job by hand. Even when the automation is accurate, removing a human's ability to review and question it treats efficiency as more important than the people affected by the change.

#### ***Data privacy***

The original export feature included client-identifying references (business names, notes) in every export, regardless of who requested it or why. That is more exposure than most requests actually need, and it gave no one a reason to pause before sharing sensitive business information further than necessary.

#### ***Explainability and trust***

The original AI suggestions were presented as a bare number with no explanation. A client trusting the system with their business should be able to understand why a recommendation was made, not just see a result and decide whether to trust it blindly.

#### ***Accountability***

Without any login system or audit trail, there was no way to know who approved a change, which conflicts with a basic ethical expectation in professional software: that consequential actions are traceable to a responsible person.

### **4. Refinements Made**

Each refinement below responds directly to one of the challenges or ethical concerns identified above. All of these are implemented in the attached code (app.py), not just described.

- Human-in-the-loop approval queue: AI-generated reorder suggestions are stored separately from live inventory and require a logged-in admin to explicitly approve or reject them before any stock quantity changes. This directly addresses the over-automation concern — the system assists a person's judgment instead of replacing it.
- Role-based access control: the system now has "staff" and "admin" roles. Both can view inventory and run the AI analysis, but only an admin can approve suggestions or export identifying data. This creates accountability for the actions that matter most.
- Full audit logging: every login attempt, AI analysis run, and suggestion approval or rejection is recorded with a timestamp and the responsible username. This solves the original traceability gap directly.
- Explainable, rule-based suggestions: rather than presenting a bare number, each AI suggestion includes a plain-language rationale (why it was suggested) and a confidence level (High/Medium/Low), so a business owner can evaluate the reasoning, not just the output.
- Privacy-conscious export: exports are anonymized (no client-identifying reference) by default for every user. An admin can choose to include full identifying detail, but only after typing a required justification, which is logged. This adds a deliberate pause before sensitive data leaves the system.
- Security hardening: all database queries use parameterized statements, passwords are hashed with Werkzeug's security helpers rather than stored in plain text, and server-side validation checks decisions (such as role and suggestion status) before any state change is applied.

- Accessibility improvements: form fields now have associated labels, flash messages are placed in an ARIA live region so status changes are announced to assistive technology, and low-stock rows are visually distinguished in the dashboard table.
- Pagination for scalability: the inventory dashboard now loads items in pages of five rather than the full table at once, as a starting point toward handling larger catalogs; the README documents the further steps (database migration, background job processing) that a production system would need beyond this classroom demo.

### **5. Decision-Making Process and Impact**

The most important decision in this refinement was resisting the temptation to make the AI feature more automated rather than more accountable. It would have been technically simpler to let approved-confidence suggestions apply themselves automatically, and that version would have looked more impressive in a demo. I chose the human-in-the-loop design instead because it is the more honest response to the ethical concern I identified: that AI-driven efficiency in my own field can come at the cost of someone's role, unless a person deliberately stays in the loop.

The impact of these changes is that the system is slower to use in the sense that nothing happens without a click of approval, but it is more trustworthy: a client can now see exactly who approved a change and why, exactly why the AI suggested what it did, and exactly what data left the system in an export. None of these changes required abandoning the original project's purpose — they required treating the people affected by the software (both the business owner and, indirectly, their staff) as more than an efficiency metric.

### **6. Summary**

This refinement did not change what the project does; it changed who stays accountable and informed while it does it. The technical additions (authentication, audit logging,

parameterized queries, pagination) and the ethical additions (human approval, explainability, privacy-conscious export) turned out to overlap more than I expected — most of the ethical fixes required a technical safeguard to actually work, and most of the technical safeguards only mattered because of the ethical concern behind them.